BUBR1-RESCUE 2.0

Genotype-Guided Drug Repurposing for Mosaic Variegated Aneuploidy

A molecular-rescuability framework for restoring residual BUBR1 function

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Track	Track 2 — Drug Repurposing
Project status	Computational and literature-supported hypothesis; no original efficacy data
Central question	Can residual BUBR1 function be pharmacologically rescued in a genotype-informed manner?
Core principle	Start from genotype and molecular defect, then match an approved drug mechanism.

Privacy note: only targeted, non-reconstructive variant summaries are reported. Raw genome-wide genotype data are excluded.

1. Executive Summary

Mosaic variegated aneuploidy (MVA) is a rare chromosome-instability disorder in which BUB1B-associated disease can result from mechanistically distinct alleles. BUBR1-RESCUE 2.0 proposes a genotype-guided drug-repurposing framework that asks whether residual BUBR1 produced by a candidate missense allele is molecularly rescuable rather than assuming that all BUB1B variants should receive the same intervention.

Targeted review of the challenge WGS identified two high-quality heterozygous BUB1B variants of particular interest. The first, NM_001211.6:c.2210T>G (p.Leu737Ter), is a known pathogenic truncating variant. The second, NM_001211.6:c.3006T>G (p.Asn1002Lys; N1002K), is an ultra-rare missense variant confirmed on the canonical/MANE transcript. Population data reviewed during the project showed approximately one observed allele and no homozygotes, while VarSome reported AlphaMissense 0.9229 as pathogenic-supporting. However, phase between the two variants and the functional consequence of N1002K remain unresolved. We therefore treat N1002K as a candidate second allele, not as a confirmed pathogenic variant.

Our central hypothesis is that N1002K may produce residual BUBR1 whose abundance or stability is impaired but whose checkpoint function could remain at least partly recoverable. This hypothesis is motivated by published MVA-associated BUB1B missense alleles, including I909T and L1012P, in which accelerated turnover and low abundance contributed to dysfunction and restoration of protein abundance could restore checkpoint activity. The proposed strategy tests two mechanistically distinct rescue routes: arimoclomol as the lead proteostasis hypothesis and nicotinic acid as an orthogonal NAD+/SIRT2-mediated BUBR1-stabilization hypothesis. Sodium phenylbutyrate (4-PBA) serves as a proteostasis comparator.

Advancement requires concordant molecular and functional rescue: increased full-length BUBR1 alone is insufficient. A candidate must show target engagement, correct BUBR1 behavior at mitosis, improved spindle-assembly-checkpoint function, reduced chromosome missegregation, and an acceptable safety signal. This creates a falsifiable Go/No-Go framework and prevents a drug from advancing solely because it increases protein abundance.

2. Clinical and Biological Problem

MVA is characterized by constitutional mosaic aneuploidies and can include growth abnormalities, developmental manifestations, and cancer predisposition. BUB1B encodes BUBR1, a core component of the spindle assembly checkpoint (SAC), which delays anaphase until chromosomes are correctly attached to the mitotic spindle.

A key translational problem is that 'BUB1B-associated MVA' does not specify a single molecular defect. Published functional work demonstrates that different BUBR1 mutations can fail through different mechanisms. Some alleles primarily reduce protein abundance or stability, whereas others retain normal abundance but have intrinsic functional defects. A rational repurposing strategy should therefore classify the defect before selecting the intervention.

3. Genomic Evidence from the Challenge Case

3.1 Targeted findings

Variant	Protein	Zygosity	QC	Interpretation	Population evidence	Unresolved issue
NM_001211.6:c.2210T>G	p.Leu737Ter	Heterozygous	PASS; GQ 99; DP 46; AD 21/25	Known pathogenic truncating BUB1B allele	Not used as the principal rarity argument	Second disease allele / phase
NM_001211.6:c.3006T>G	p.Asn1002Lys (N1002K)	Heterozygous	PASS; GQ 99; DP 28; AD 15/13	Candidate missense second allele; not established pathogenic	Ultra-rare; ~6.2×10^{-7} total AF reviewed; 0 homozygotes	Functional consequence and phase

3.2 Interpretation boundaries

The p.Leu737Ter call is independently supported by ClinVar as pathogenic for MVA1. N1002K is a missense change in the C-terminal BUBR1 region and was independently annotated by Ensembl VEP as a canonical-transcript missense variant. Computational evidence increases its priority for follow-up but does not establish causality.

Because the VCF genotypes are unphased and no phase-set information was present for these calls, the project does not claim that p.Leu737Ter and N1002K are in trans. Accordingly, the phrase 'compound heterozygous causal genotype' is intentionally avoided.

4. Genotype-to-Mechanism Interpretation

Published MVA studies provide a useful mechanistic precedent. Suijkerbuijk and colleagues showed that selected disease-associated BUBR1 missense mutants can exhibit increased turnover and low protein abundance. I909T and L1012P were particularly informative: their instability was linked to chaperone/proteostasis dependence, and restoring sufficient mutant protein abundance could restore checkpoint activity. This establishes that at least some BUBR1 missense alleles are abundance-limited rather than completely function-null.

N1002K is not experimentally characterized in those studies, so behavior cannot be transferred from L1012P or I909T by proximity alone. Instead, these alleles motivate a testable classification question: does N1002K behave as an abundance/stability defect, an intrinsic functional defect, or neither?

5. Central Hypothesis: Molecular Rescuability

Hypothesis. If N1002K produces residual BUBR1 that is destabilized or excessively degraded but retains recoverable checkpoint activity, then a mechanism-matched approved drug may increase functional full-length BUBR1 and improve chromosome-segregation phenotypes.

This hypothesis is deliberately conditional. If N1002K is normally abundant but intrinsically defective, if it is functionally neutral, or if it is not the second disease allele, the stabilization strategy should not advance.

6. Drug Repurposing Strategy

6.1 Lead hypothesis: Arimoclomol — proteostasis rescue

Arimoclomol is prioritized as the lead genotype-conditioned experiment because published BUBR1 mutant biology makes proteostasis a plausible vulnerability for selected MVA missense alleles. Its role here is not based on direct evidence that arimoclomol rescues BUBR1; no such evidence is claimed. Rather, it is a mechanistic test of whether enhancing the cellular stress/proteostasis response can increase functional residual BUBR1 from the candidate missense allele.

Translationally, arimoclomol has an important repurposing advantage: the FDA approved Miplyffa (arimoclomol) in September 2024, in combination with miglustat, for neurological manifestations of Niemann-Pick disease type C in adults and children two years of age and older. This approval demonstrates human clinical development and regulatory precedent, but does not imply efficacy or safety for MVA.

6.2 Orthogonal hypothesis: Nicotinic acid — NAD+/SIRT2/BUBR1 stabilization

Nicotinic acid provides a mechanistically independent route. Hara and colleagues demonstrated that nicotinic acid can raise cellular NAD in human cells through NAPRT. Separately, North and colleagues showed that SIRT2 regulates BUBR1 abundance: CBP-mediated acetylation at K668 promotes BUBR1 degradation, whereas SIRT2-mediated deacetylation stabilizes BUBR1; increasing NAD with NMN increased BUBR1 abundance in vivo.

The unproven bridge is central to this project: nicotinic acid has not been shown to rescue BUBR1 in human MVA cells, and it is unknown whether N1002K would respond to this pathway. Therefore, nicotinic acid is an orthogonal mechanistic candidate rather than an asserted therapy.

6.3 Comparator: Sodium phenylbutyrate (4-PBA)

4-PBA is retained as a chemical-proteostasis comparator with repurposing precedent. It is not ranked as the primary lead because the current evidence connecting it to BUBR1 is weaker than the allele-specific proteostasis rationale motivating the arimoclomol experiment and weaker than the direct BUBR1 stability biology underlying the NAD+/SIRT2 route.

6.4 Mechanism-based prioritization

Candidate	Primary mechanism	Direct BUBR1	N1002K match	Role

	tested	**evidence**		
Arimoclomol	Proteostasis / stress-response enhancement	Indirect; motivated by unstable BUBR1 mutant biology	High-value hypothesis, unproven	Lead
Nicotinic acid	NAPRT → NAD+ → SIRT2-linked BUBR1 stabilization	Direct pathway evidence for BUBR1 stability; not direct for niacin in MVA	Unknown	Orthogonal candidate
4-PBA	Chemical proteostasis support	No direct BUBR1 rescue evidence	Unknown	Comparator

7. Evidence Ladder

Level	Question	Current status	What advances confidence
1. Genomic	Is there a plausible BUB1B genotype?	One known pathogenic truncating allele + one ultra-rare missense candidate	Phase/segregation and independent confirmation as available
2. Variant mechanism	Does N1002K reduce abundance/stability?	Unknown	BUB1B mRNA vs full-length BUBR1 protein; turnover/proteostasis assays
3. Drug target engagement	Does the drug engage its proposed route?	Untested	Arimoclomol: stress/proteostasis response; niacin: NAPRT-dependent NAD+ increase
4. Molecular rescue	Does full-length BUBR1 increase appropriately?	Untested	Protein abundance plus localization
5. Functional rescue	Does SAC/chromosome segregation improve?	Untested	Checkpoint and chromosome-missegregation endpoints
6. Safety	Does rescue avoid adverse proliferative/tumor-fitness effects?	Untested	Cell-fitness and safety gate

8. Experimental Validation Plan

8.1 Stage 1 — Establish the molecular defect

Measure BUB1B transcript and full-length BUBR1 protein in a patient-relevant cellular model or a controlled N1002K model. A normal transcript with reduced protein would support a post-transcriptional abundance/stability defect. Normal protein abundance with defective checkpoint behavior would argue against a simple stabilization strategy.

8.2 Stage 2 — Mechanistic drug comparison

Compare vehicle, arimoclomol, nicotinic acid, and 4-PBA under matched experimental conditions. The first molecular endpoint is full-length BUBR1 abundance. Drug-specific target-engagement measurements should accompany abundance: NAD and NAPRT context for nicotinic acid, and an appropriate heat-shock/proteostasis-response readout for arimoclomol.

8.3 Stage 3 — Functional rescue

Only conditions showing credible molecular rescue advance to function. Evaluate BUBR1 localization at kinetochores, spindle-assembly-checkpoint competence, and chromosome-segregation outcomes such as lagging chromosomes, micronuclei, or aneuploidy-related readouts. The exact assay panel can be adapted to the available model, but the logical chain must remain intact.

8.4 Stage 4 — Safety gate

Because MVA is associated with cancer predisposition and BUBR1 is a mitotic regulator, a successful molecular rescue must not be interpreted as automatically beneficial. Advancement requires no concerning increase in inappropriate proliferative fitness or other safety signals in the experimental system.

9. Go/No-Go Decision Framework

Checkpoint	GO	NO-GO	Interpretation
Baseline defect	Reduced full-length BUBR1 with evidence compatible with residual function	Normal abundance with clear intrinsic dysfunction, or no relevant N1002K phenotype	Determines whether stabilization is rational
Target engagement	Drug changes its intended proximal pathway	No proximal pathway response	Do not attribute downstream changes to proposed mechanism
BUBR1 rescue	Full-length BUBR1 rises appropriately	No increase, or only nonfunctional/aberrant product	Abundance alone is not sufficient
SAC/segregation	Checkpoint and	No functional	Reject rescue

	chromosome-segregation phenotypes improve	improvement despite increased BUBR1	hypothesis for that condition
Safety	No concerning fitness/safety signal	Adverse proliferative or tumor-fitness signal	Stop advancement

10. Mechanism Router

The framework is designed to survive a negative result. If N1002K is an abundance/stability defect with recoverable function, proteostasis and stabilization strategies advance. If N1002K is normally abundant but intrinsically defective, stabilization is a No-Go and the project redirects toward function-specific mechanisms. If p.Leu737Ter and N1002K are found to be in cis, the proposed biallelic model is unresolved and the analysis returns to second-allele discovery, including structural/CNV possibilities if supported by the available dataset.

11. Innovation

The principal innovation is not a single drug-disease pairing. BUBR1-RESCUE converts variant interpretation into a drug-selection problem by asking whether a specific allele is molecularly rescuable. This reduces the risk of treating all variants in the same gene as mechanistically equivalent.

A second innovation is orthogonal rescue testing. Arimoclomol and nicotinic acid interrogate different molecular bottlenecks—proteostasis versus BUBR1 stability through an NAD+/SIRT2-linked axis—while converging on the same requirement: restoration of functional BUBR1.

12. Potential Impact

If validated, the framework could identify a tractable therapeutic hypothesis for a subset of BUB1B-associated MVA in which residual mutant protein retains rescuable function. Even a negative result would be informative because it would prevent advancement of a stabilization strategy for a function-null allele.

The approach also demonstrates how controlled genomic evidence, mechanistic literature, approved-drug biology, and explicit falsification criteria can be integrated in an ultra-rare disease setting where conventional clinical datasets are necessarily small.

13. Scalability

The general workflow is transferable beyond BUB1B: (1) identify a candidate genotype, (2) classify the molecular defect, (3) determine whether residual protein is rescuable, (4) match approved drugs to the defect, and (5) require target engagement plus phenotypic rescue before advancement. This is more scalable than a disease-name-to-drug lookup because it operates at the level of variant mechanism.

14. Limitations and Falsification

- Phase between p.Leu737Ter and N1002K is unresolved; a biallelic molecular diagnosis is not claimed.
- N1002K has no direct functional characterization in this project and is treated as a candidate VUS/second allele rather than a confirmed pathogenic variant.
- AlphaMissense and other computational predictors are supportive evidence only; they cannot establish pathogenicity.
- Proximity to previously characterized MVA alleles such as L1012P does not prove that N1002K shares their mechanism.
- No original drug-efficacy experiments have been performed.
- Arimoclomol has not been shown to rescue BUBR1 or MVA.
- Nicotinic acid has not been shown to rescue BUBR1 in human MVA cells; the NAPRT→NAD and SIRT2→BUBR1 findings come from separate evidence streams.
- An increase in BUBR1 abundance without restoration of checkpoint/chromosome-segregation function is a negative result, not therapeutic validation.
- Any translational interpretation must account for MVA-associated cancer predisposition and drug-specific safety.

15. Conclusion

BUBR1-RESCUE 2.0 reframes drug repurposing for this MVA case around molecular rescuability. The challenge WGS supports a known pathogenic BUB1B truncating allele and a second ultra-rare N1002K missense candidate, while leaving phase and N1002K pathogenicity unresolved. Published BUBR1 biology nevertheless provides a strong reason to test whether the candidate missense allele is abundance-limited and recoverable. Arimoclomol is therefore prioritized as the lead proteostasis experiment, nicotinic acid as an orthogonal BUBR1-stabilization experiment, and 4-PBA as a comparator. The proposal succeeds scientifically only if molecular target engagement leads to restoration of functional full-length BUBR1, improved mitotic checkpoint behavior, reduced chromosome missegregation, and an acceptable safety profile.

Judge-facing interpretation. The project is optimized for the four Track 2 criteria: scientific rigor through explicit uncertainty and falsification; potential impact through a patient-specific but mechanism-based rescue strategy; innovation through molecular-rescuability routing; and scalability through a reusable genotype-to-mechanism-to-drug workflow.

Reproducibility. The accompanying GitHub package contains the evidence table, claim-discipline rules, analysis scripts/configuration needed to reproduce the candidate prioritization logic, and documentation of the manual annotation steps. Controlled raw genomic files are intentionally excluded.

AI assistance disclosure. OpenAI ChatGPT, Plus plan, was used to assist with literature organization, hypothesis structuring, drafting, and code/document preparation. The account-level training/data-sharing setting was not recorded for this project. To minimize exposure of controlled genomic data, raw FASTQ and full VCF files were not uploaded to the assistant; only targeted, non-reconstructive variant summaries and selected annotation outputs were used.

Track 2 fit. This submission proposes approved-drug repurposing hypotheses for follow-up and does not claim therapeutic efficacy. The mechanism characterization is explicitly conditional: the known p.Leu737Ter allele is loss-of-function/truncating, while N1002K is treated as an unresolved missense candidate whose molecular defect must be experimentally classified.

16. Hackathon Alignment and Methods Transparency

17. References

1. Suijkerbuijk, S. J. E., et al. (2010). Molecular causes for BUBR1 dysfunction in the human cancer predisposition syndrome mosaic variegated aneuploidy. Cancer Research, 70(12), 4891–4900. https://doi.org/10.1158/0008-5472.CAN-09-4319
2. North, B. J., et al. (2014). SIRT2 induces the checkpoint kinase BubR1 to increase lifespan. The EMBO Journal, 33(13), 1438–1453. https://doi.org/10.15252/embj.201386907
3. Hara, N., Yamada, K., Shibata, T., Osago, H., Hashimoto, T., & Tsuchiya, M. (2007). Elevation of cellular NAD levels by nicotinic acid and involvement of nicotinic acid phosphoribosyltransferase in human cells. Journal of Biological Chemistry, 282(34), 24574–24582. https://doi.org/10.1074/jbc.M610357200
4. National Center for Biotechnology Information. ClinVar record for NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter), Mosaic variegated aneuploidy syndrome 1. Accessed August 2026.
5. U.S. Food and Drug Administration. (2024, September 20). FDA approves first treatment for Niemann-Pick disease, type C (Miplyffa/arimoclomol).
6. Ensembl Variant Effect Predictor (VEP). Annotation of BUB1B NM_001211.6:c.3006T>G, p.Asn1002Lys, GRCh38. Project analysis, 2026.
7. gnomAD v4.1.1. Population frequency review for GRCh38 15-40220612-T-G. Project analysis, 2026.
8. VarSome. Germline annotation for GRCh38 15-40220612-T-G; AlphaMissense 0.9229, pathogenic-supporting. Project analysis, 2026.

Appendix A. Claim Discipline

Supported wording	Avoid
Known pathogenic heterozygous p.Leu737Ter allele	Confirmed complete molecular diagnosis
Ultra-rare N1002K candidate second allele	N1002K is pathogenic
Phase unresolved	Compound heterozygous
Arimoclomol is a genotype-conditioned rescue	Arimoclomol treats MVA

hypothesis	
Nicotinic acid is an orthogonal mechanistic candidate	Niacin rescues BUBR1 in MVA
Published neighboring/related alleles motivate testing	N1002K behaves like L1012P